

NEXGO Android Smart PinPAD Programmers Manual

Detailed host command fields, response formats, and operating flows

Revision 1.1 - July 3, 2026

1. Scope

This manual describes the NEXGO Android Smart PinPAD host protocol implemented by the PINPAD application. The device performs L2 card interaction, secure PIN entry, display prompts, local key management, and card-reader control. It does not perform host authorization or settlement.

- The host talks to the device through RS232 or USB CDC virtual serial transport.
- The main screen remains an idle prompt unless a command changes the display state.
- All customer-facing text is localized by the Android application or by the configured prompt text.
- Configuration commands persist EMV data in application storage and reload kernel configuration when needed.

2. Packet and Session Rules

Packet Controls

Item	Definition	Notes
Transaction frame	<STX> Command Payload <ETX> LRC. Used for card, display, EMV, PIN, JPEG, CPU, and RF commands.	The host must ACK or NAK device responses. Transaction responses do not send a final EOT after host ACK.
Administration frame	<SI> Command Payload <SO> LRC. Used for device administration and key status commands.	Administration responses send final EOT after host ACK, except commands 11 and 14.
Field separator	<FS> hexadecimal 1C.	Separates variable fields inside request and response payloads.
Subfield delimiter	<SUB> hexadecimal 1A.	Used by prompts, transaction display, multipacket payloads, and some EMV responses.
Bad packet	Valid LRC but unsupported message id is ignored; valid id with wrong format returns EOT or a command-specific fail response.	Partial frames time out and are discarded. Garbage is ignored until a new frame/control start byte is seen.
Device response retry	If host does not ACK/NAK a device packet within the retry window, the response is retransmitted. Repeated failure terminates the session with EOT.	NAK from the host causes retransmission of the last packet.

3. Shared Field Conventions

Common Data Types and Result Codes

Item	Definition	Notes
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Item	Definition	Notes
Status 0/1	0 means accepted or completed. 1 means failed. Some commands append a reason code after 1.	EMV setup failures use 1 + reason + optional error text.
Key id	Normal master-key ids are 0-9 and A-G. Secret PIN session ids are 0-9 and map internally to secure slots 20-29.	Secret session keys wait for the secret master key to finish loading before injection.
Key usage	K0 master key; P0 PIN working key; M3 MAC key; D0 data encryption key; K1 reserved for key-loading/key-encryption use.	P0 is stored as a working PIN key under a random wrapper master key.
EMV result	Y1 offline approved, Z1 offline declined, Y3 unable online/offline approved, Z3 unable online/offline declined, Y4 online approved, Z4 online declined, A1 online request, A4 application reselection.	Declined results show a declined screen and error beep. Approved sale results may show brand sensory if enabled.
Transaction display	ZA stores transaction type, amount, currency, and optional symbol for the next QK, Q1, T11, T61, or related read flow.	Cleared by Z1, 72, T1C, T6C, physical cancel, or transaction-state reset.
Prompt ids	Fixed prompt identifiers include PIN prompts and data-entry prompts. Custom text can also be sent directly.	Available prompt ids are listed in the prompt appendix.

4. Detailed Command Dictionary

4.1 Administration and Device Commands

Command	Request fields	Response fields	Flow and notes
02 Load Key	[KeyId][KeyMaterial][<FS>UsageModeAlgorithm?] KeyId: 0-9,A-G. Material: clear key hex. Optional attribute: usage(2)+mode+algorithm.	Success: echoes request frame. Error: 02?x where x is error code. Final EOT after host ACK.	Loads normal MK/SK master and working-key material. K0 writes an SDK master key. P0/M3/D0 are stored as working PIN/MAC/data keys. Invalid format returns EOT after frame ACK policy.
04 Check Master Key	[KeyId][AttrFlag?] AttrFlag 1 asks for key attributes.	04F if loaded, 040 if missing. With AttrFlag 1: 04F[Usage]<FS>[Mode]<FS>[Algorithm].	Queries loaded key state and locally stored key metadata. If metadata is missing after reinstall, PIN operations fail and instruct the user to inject keys again.
05 Load Serial Number	[SerialNumber up to 16 chars]	Echoes 05 with accepted serial substring.	Compatibility command only. Device hardware serial number is not changed.
06 Get Serial Number	No payload.	06[DeviceSerialNumber].	Returns the Android/NEXGO device serial number available from the SDK/device properties.
08 Select Active Master Key	[KeyId]	080 for success or 081 for failure.	Sets active MK/SK key for PIN and MAC operations. Uses local key metadata to validate later PIN scheme behavior.
09 Communications Test	Step 1: 09. Step 2 host echoes 09<SUB>PROCESSING.	Step 1 response: 09<SUB>PROCESSING. Step 2 response: 090. Bad echo returns EOT.	Two-step communication loopback. Final EOT follows the accepted administration response.
11 Device Connection Test	No payload.	ACK only.	Connection probe. Does not send final EOT.

Command	Request fields	Response fields	Flow and notes
12 Prompt Language	[LanguageIndex]	120 accepted or 121 rejected.	Compatibility response. Android system/application locale remains authoritative for the app UI.
13 Baud/Port Speed	[BaudCode][Mode?]	130 accepted or 131 rejected.	Validates baud code 1-8 and optional mode. Actual serial settings are controlled by setup configuration.
14 Acknowledge	No payload.	ACK only.	Compatibility no-op. Does not send final EOT.
17 Random Number	No payload.	17[8-byte random hex].	Returns a random 8-byte block generated by device crypto service or fallback random source.
19 Firmware Version	[Component][Option?]	19.[Version][.SubVersion?][.Hash?].	Versions are prefixed with NEXGO-model-compatible text used by host tools. Component 4 returns the application version.
1C Hardware Capability	No payload.	1C<FS>[Capability] repeated.	Reports supported reader/PIN/display capability strings.
1F Prompt Table Query	No payload.	1Fes<FS>en<FS>es.	Reports supported prompt language table identifiers.
1M Keypad Beeper Setup	0 disable, 1 enable.	1M0 or 1M1 on success; 1M0 on invalid/failure.	Controls keypad beep preference.
1P Control Beeper	<SUB>[Count]<FS>[DurationUnits]<FS>[IntervalUnits]	1P0 success or 1P1 failure.	Beeps Count 1-9. Duration and interval are in 10 ms units.
M03 Load Permanent Serial	[11-char serial][Slot?]	M030 success, M031 bad serial, M032 already set, M033 storage/slot error.	Stores a protected permanent unit serial in reinstall-persistent storage. Slot defaults to 0.
M04 Query Permanent Serial	[Slot?]	M04[Serial] or M04 empty if never loaded. M041 invalid slot.	Returns the stored permanent unit serial. Must return empty when no value has been loaded.

4.2 Key Management and PIN Entry

Command	Request fields	Response fields	Flow and notes
20 Load Secret Master Key	[Option][KeyHex] Option 0/1. KeyHex 16 or 32 hex chars.	Success: echoes command 20. Format/SDK error: EOT.	Accepts secret master key and queues/injects it for secret slots 20-29. Visual cue: Secret Master Key Loaded. Used by commands 21-24.
21 Load Secret Session Key	[KeyId][EncryptedSessionKeyHex] KeyId 0-9. Key 16 or 32 hex chars.	Success: echoes command 21. Format/SDK error: EOT.	Loads encrypted PIN session key into secret slot 20+KeyId after the secret master key exists. Visual cue: Secret Session Key Loaded.
22 Secret PIN Entry	[PAN]<FS>[KeyId...]	71[PIN result] or EOT on cancel/timeout.	Starts secure PIN entry using the secret MK/SK session key slot. Default PIN length 4-12.
23 Secret PIN Entry External	[PAN]<FS>[KeyId...]	71[PIN result] or EOT.	Same secure flow as 22, with host-provided prompt behavior where supported.

Command	Request fields	Response fields	Flow and notes
Prompt			
24 Secret PIN Entry Custom Prompt	[PAN]<FS>[KeyId][MinMaxNullPrompt]<FS>[Prompt2?]<FS>[Completion?]	71[PIN result] or EOT.	Uses custom PIN control: min two digits, max two digits, Y/N null flag, then prompt text.
60 DUKPT PIN Entry	[PAN/Account data]	71[Status/PINBlock/KSN] or immediate error.	Starts DUKPT PIN entry. Account is validated for PIN block formatting.
62 Amount Authorization	Compatibility payload.	63[0].	Auto-confirms amount authorization for compatibility.
63 PIN Response From Host	Response-id compatibility frame.	ACK only.	Accepted as no-op when received from host.
70 PIN Entry	DUKPT form or .[PAN]<FS>[SessionKey/control].	71[PIN result] or EOT.	If payload starts with dot, uses MK/SK; otherwise DUKPT. MK/SK requires active key metadata.
71 PIN Result Response Id	Response-id compatibility frame.	ACK/no-op.	Device sends 71 for PIN result. Host should not send it except compatibility no-op.
7G PIN Change	.[PAN]<FS>[SessionKey/control].	71[PIN result] or EOT.	Starts MK/SK PIN change style entry.
7A DUKPT KSN Format	[FormatFlag]	ACK only.	Controls full/stripped KSN output format for DUKPT PIN responses.
90 Load DUKPT IPEK Set 0	[IPEK32][KSN20]	91[0] success or 91[1][Reason].	Injects DUKPT initial key for set 0. Visual cue: IPEK Loaded on success.
94 Load DUKPT IPEK Set 1	[IPEK32][KSN20]	91[0] success or 91[1][Reason].	Injects DUKPT initial key for set 1.
96 Select DUKPT Key Set	[0 or 1]	ACK only.	Selects active DUKPT key set.
98 Query DUKPT Key Set	[KeySet][IncludeInfo?]	99[Status] or 99F B1<FS>X<FS>T<FS>[KCV].	Returns key state. IncludeInfo 1 appends algorithm/type/KCV style fields.
99 DUKPT Status Response Id	Response-id compatibility frame.	ACK/no-op.	Device sends this as response to 98 or load-key status operations.
Z60 PIN Entry External Prompt	MK/SK or DUKPT PIN request with prompt selection.	71[PIN result] or EOT.	Uses fixed/external prompt id and PIN request controls.
Z62 PIN Entry Custom Prompt	[PAN]<FS>[SessionKey?][MinPin][MaxPin][NullFlag][Prompt1]<FS>[Prompt2]<FS>[Completion]<FS>[Timeout?]	71[PIN result] or EOT.	Displays custom PIN prompts. Shows a masked echo box and responds on enter, cancel, timeout, or SDK result.
Z64 Query Key KCV	[KeyId]	Z65[KeyId][KCV] or Z65[KeyId]?	Calculates KCV for a loaded master key. Unknown/missing key returns question marker.

Command	Request fields	Response fields	Flow and notes
Z65 KCV Response Id	Response-id compatibility frame.	ACK/no-op.	Device response to Z64.
Z66 Generate MAC	MAC input payload.	Z670[MAC16] success or Z672 failure.	Uses active MAC-capable key metadata and SDK MAC function.
Z67 MAC Response Id	Response-id compatibility frame.	ACK/no-op.	Device response to Z66.

4.3 Magnetic Stripe, Track-Equivalent, and Multi-Interface

Command	Request fields	Response fields	Flow and notes
Q1 Arm MSR	No payload.	Asynchronous 81[TrackData] after swipe; EOT on cancel.	Displays swipe prompt/animation. If QK cached an MSR swipe, Q1 consumes cached data instead of rearming.
Q2 Transaction Complete	No payload.	ACK/no frame.	Clears card-read state and displays thank-you then idle.
Q3 Ignore Card Swipe	No payload.	ACK/no frame.	Stops MSR/card detection and shows processing/idle state.
Q4 Set MSR Track Mode	[Flag] 0 track2 only, 1 disabled, 2 all tracks.	ACK/no frame.	Persists MSR track mode. Disabled mode stops reader search.
Q5 Set MSR Retry Count	[0-9]	ACK/no frame.	Stores retry count for swipe handling.
Q6 Set MSR Output Format	[0 or 2]	Echo Q6 on success; EOT on bad format.	0 clear text without sentinels; 2 clear text with sentinels.
Q7 Query MSR Format	No payload.	Q7[FormatFlag].	Returns current MSR output format.
Q8 PCD Track- Equivalent Read	No payload.	83[TrackData] or EOT.	Starts RF card read that returns track-equivalent data from EMV tags.
Q9 Dual MSR/PCD Read	No payload.	81[TrackData] for swipe or 83[TrackData] for RF.	Arms both swipe and RF paths; first successful reader wins.
QA Set PCD Track Mode	[Flag] 0 track2 only, 1 disabled, 2 all tracks.	ACK/no frame.	Controls contactless track-equivalent mode.
QB MSR Auto Arm	[0 disable, 1 enable]	QB0 success or QB1 failure.	When enabled, MSR reader remains armed and sends 81 on swipe.
QC PCD Auto Arm	[0 disable, 1 enable]	QC0 success or QC1 failure.	Controls contactless auto-arm compatibility flag.
QD Set PCD	[0 or 2]	Echo QD on success; EOT on invalid.	Sets contactless track-equivalent output format.

Command	Request fields	Response fields	Flow and notes
Output Format			
QF MSR With JPEG Prompt	Prompt/image payload ignored for display.	Asynchronous 81[TrackData] or EOT.	Uses current Android animation rather than legacy image assets.
QG PCD With JPEG Prompt	Prompt/image payload ignored for display.	Asynchronous 83[TrackData] or EOT.	Uses contactless read UI and soft LEDs.
QH Dual Read With JPEG Prompt	Prompt/image payload ignored for display.	81 or 83 track response.	Dual-reader version of Q9.
QI Query Contactless Tags	[ReadyControl?][DOL/tag list]	QJ[F] if no data or QJ11<FS>[TLV data].	Reads currently cached/kernel EMV tags for contactless card data.
QK Multi-Interface Detect	No payload.	QK1 swipe, QK2 chip, QK3 contactless, QK0x error/cancel.	Displays swipe/insert/tap prompt and caches track data for Q1/Q8 when possible. Cancel sends QK03/EOT behavior per active operation.
81 MSR Data Response	Device-to-host only.	81[Track1]<FS>[Track2]<FS>[Track3] formatted by Q6/Q4.	Sent asynchronously after Q1/Q9/QF or auto-arm swipe.
83 PCD Track Response	Device-to-host only.	83[Track1]<FS>[Track2]<FS>[Track3].	Sent after Q8/Q9/QG/QH or cached contactless data extraction.

4.4 CPU Smart Card Commands

Command	Request fields	Response fields	Flow and notes
I00 Primary Card Present	No payload.	I00F present or I000 absent.	Checks ICC1 card presence.
I01 Primary Cold Reset	No payload.	I02[ATR] or I0F[Error].	Powers on ICC1 and returns ATR.
I02 Primary ATR Response Id	Host request treated unsupported.	I0F09.	Device uses I02 as response to I01.
I03 Unsupported CPU Command	Any payload.	I0F09.	Accepted message id; no SDK operation implemented.
I04 Primary Deactivate	No payload.	I04 or I0F08.	Powers off ICC1.
I05 Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.

Command	Request fields	Response fields	Flow and notes
I06 Primary APDU Exchange	[APDU hex 4-262 bytes].	I07[ResponseAPDU] or I0F[Error].	Requires a prior I01 reset/power-on.
I07 Primary APDU Response Id	Host request treated unsupported.	I0F09.	Device uses I07 as response to I06.
I08 Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I09 Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I0A Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I0B Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I0C Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I0D Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I0E Unsupported CPU Command	Any payload.	I0F09.	Reserved compatibility command.
I0F CPU Error Response Id	Host request treated unsupported.	I0F09.	Error codes: 01 no/reset fail, 02 length, 03 not powered, 04 bad hex, 06 bad interface, 08 SDK failure, 09 unsupported, 0A no device.
I11 SAM Cold Reset	No payload.	I12[ATR] or I0F[Error].	Powers on selected SAM slot. Default active slot is PSAM1 unless changed by I15.
I12 SAM ATR Response Id	Host request treated unsupported.	I0F09.	Device uses I12 as response to I11.
I14 SAM Deactivate	No payload.	I14 or I0F08.	Powers off selected SAM.
I15 Select SAM Interface	[1 PSAM1, 2 PSAM2].	I15[Interface] or I0F06.	Changes active SAM slot and clears SAM powered state.
I16 SAM APDU Exchange	[APDU hex].	I17[ResponseAPDU] or I0F[Error].	Requires prior I11 power-on for selected SAM.

Command	Request fields	Response fields	Flow and notes
I17 SAM APDU Response Id	Host request treated unsupported.	I0F09.	Device uses I17 as response to I16.

4.5 MIFARE and RF Memory Card Commands

Command	Request fields	Response fields	Flow and notes
P01 RF Enable/Disable	[0 disable, 1 enable].	P010 success or P011 format/operation error.	Opens/closes RF reader and clears cached RF card state when disabled.
P02 Query Presence	No payload.	P020[ATQA] or P013.	Refreshes Type A info and returns ATQA.
P03 Anticollision	No payload.	P030[UID] or P013.	Returns UID after Type A anticollision-style read.
P04 Select	No payload.	P040[SAK] or P013.	Returns SAK.
P05 Activate	No payload.	P050[ATQA]<FS>[SAK]<FS>[UID] or P013.	Returns activation data.
P06 Halt	No payload.	P060 or P013.	Closes/reopens RF reader to halt/deselect current card.
P07 Authenticate Classic	[Sector2] or [Sector2][A/B][Key12] or [Sector2][KeyNo3][A/B].	P070 or P011/P013.	Authenticates first block of sector with direct key or stored key from P15.
P08 Read Ultralight Page	[Page2].	P080[4-byte hex] or fail.	Reads page 0-47 from Ultralight-family card.
P09 Write Ultralight Page	[Page2][Data8].	P090 or fail.	Writes 4 bytes to page 0-47.
P10 Read Block	[Block3].	P100[BlockDataHex] or fail.	Classic reads 16 bytes; Ultralight-family path reads page data.
P11 Write Block	[Block3][Data32].	P110 or fail.	Classic writes 16 bytes; Ultralight path writes first 4 bytes.
P12 Read Sector	[Sector2].	P120[SectorDataHex] or fail.	Reads all blocks in sector. Sectors 0-31 have 4 blocks; 32-39 have 16 blocks.
P13 Write Sector	[Sector2][DataHex multiples of 32].	P130 or fail.	Writes sequential blocks within the sector.
P14 Value Operation	[Block3][Mode][Value/Target fields].	P140 or P14[1][Reason].	Mode 0 write value, 1 decrement, 2 increment, 3 backup. Reason 3/5/6 identifies operation failure class.
P15 Load RF Key	[KeyNo3][KeyA12][KeyB12].	P150 or fail.	Stores MIFARE keys in app preferences for later P07 authentication.
P16 Identify Type	No payload.	P160-TypeCode or fail.	Type codes: 1 Ultralight, 2 Classic 1K, 3 Classic 4K, 4 DESFire, 5 Plus, 6 Mini, 0 unknown.

Command	Request fields	Response fields	Flow and notes
P17 Activate DESFire	No payload.	P170[ATS/UID] or fail.	Succeeds only when detected RF type is DESFire.
P18 Deselect DESFire	No payload.	P180 or fail.	Clears RF state and reopens reader when enabled.
P19 APDU Exchange	[APDU hex].	P190[ResponseHex] or fail.	Uses RF CPU APDU for DESFire/Type A CPU, otherwise raw RF exchange.
P20 Raw Block Exchange	[CrcMode][WaitTime4][DataHex].	P200[ResponseHex] or fail.	Performs raw RF exchange. CrcMode 0/1; WaitTime is accepted for compatibility.

4.6 Display, Prompts, Data Entry, and Images

Command	Request fields	Response fields	Flow and notes
B1 Set Font Size	[Option 0-5].	B2[Status].	Compatibility display setting. Android UI scales according to app layout.
B3 Set Font Color	[ForegroundRGB6][BackgroundRGB6].	B4[Status].	Validates RGB hex and accepts color preference.
BB Screen Saver Setup	[WaitTime3][Type].	BC[Status].	Accepts wait 005-999 and type 1. Screen saver behavior is controlled by Android/device policy.
BD Screen Saver Enable	[0 disable, 1 enable].	BE[Status].	Compatibility response.
BF Screen Saver Preview	[Operation].	ACK/no frame.	Compatibility no-op.
Z0 Move Cursor	[Line2].	ACK/no frame or EOT on bad format.	Compatibility command for text display cursor.
Z1 Reset State	No payload.	ACK/no frame.	Clears transaction display, data-entry prompts, cached card data, active PIN/card reads, and returns to idle.
72 Cancel Transaction	No payload.	ACK/no frame.	Same reset/cancel behavior as Z1; also cancels PIN/card operations.
Z2 Display Prompt	Raw text, <GS>[PromptId3], <RS>[PinPromptId3], or <FS> clear.	Optional Z2[Status] for fixed prompt form.	Displays one prompt. Fixed prompt status: 0 OK, 1 bad id, 4 clear/idle.
Z3 Display Multi-Line Prompt	Raw counted lines, <GS>id list, <RS>PIN id list, or <FS> clear.	Optional Z3[Status].	Displays up to 7 lines or fixed prompt text list. Auto returns to idle after configured message duration where applicable.
Z7 Cancel Message Option	[0 show cancel, 1 suppress cancel].	ACK/no frame or EOT on bad option.	Controls whether data-entry cancel shows Operation Cancelled.

Command	Request fields	Response fields	Flow and notes
Z8 Idle Prompt	[PromptText].	ACK/no frame.	Sets customer idle prompt. Default is NEXGO WELCOME.
ZA Transaction Display	[TxnType]<FS>[AmountMinor]<FS>[CurrencyCode]<FS>[CurrencySymbol?].	ACK/no frame; EOT on invalid payload.	Stores amount/type/currency for the next read/transaction UI. Amount is implicit decimal minor units.
Z42 Read Key Code	[TimeoutSeconds 1-255].	Async Z43[KeyCode] or EOT.	Requires a prompt already prepared by Z2/Z3. Numeric and control keys are returned according to keypad mapping.
Z43 Key Code Response	Device-to-host only.	Z43[KeyCode].	Sent after user presses a valid key during Z42.
Z50 String Entry	[EchoMode][Timeout][MaxLength] format from host payload.	Async Z51[EnteredText] or EOT.	Requires prompt ready. Enter submits, Clear clears buffer, Cancel aborts. Echo mode controls visible/masked display.
Z51 String Entry Response	Device-to-host only.	Z51[Text].	Sent when user completes Z50 entry.
J0 Initialize Image Table	No payload.	J0[0/1].	Initializes local image file table.
J1 Query Image Table	No payload.	J11<FS>[SelectedFlag+Name] repeated.	Lists image names tracked by the app.
J2 Select Images	[Control]<FS>[Name] repeated.	J2[Status] repeated by FS.	Marks images selected for playback or use.
J3 Delete Images	[Name] repeated by FS.	J3[Status] repeated by FS.	Deletes stored image files.
J4 Download Image Packet	Packet payload.	J4[Status].	Stores packet data. Current app does not use legacy 320x240 assets for primary UI.
J5 Upload Image Packet	0<FS>[Name] start or 1 next.	J5[PacketPayload].	Returns image upload chunks/status.
J6 Play Selected Images	No payload.	ACK/no frame.	Displays selected image sequence if any.
J7 Set Idle Image	[Name].	J70.	Accepted for compatibility; idle logo image is not used for main UI.
J8 Enable Idle Image	[Operation].	J80.	Accepted for compatibility; Android idle prompt remains primary.
J9 Show Image	[Name].	J9[Status].	Displays a stored image if found.
JA Download Boot Logo	Packet payload.	JA[Status].	Compatibility storage command.

4.7 EMV Contact Configuration and Contact Transactions

Command	Request fields	Response fields	Flow and notes
T01 Load Terminal Config	Configuration packet/TLV payload.	T020 success or T021[Reason][Error].	Stores terminal parameters and flags EMV configuration for reload.
T03 Load CAPK	[Sequence][CAPK header/modulus packet].	T04[Sequence][Status].	Stores CAPK. Shared SDK CAPK store is used for contact and contactless.
T05 Load Application Config	Application packet/TLV payload.	T060 success or failure.	Stores AID/application configuration for kernel reload.
T07 Load Data Formats	[ClearFlag][TagFormatDefinitions].	T080 success or failure.	Stores private tag format definitions used later for custom EMV tag output.
T09 Query Config IDs	[ConfigType].	T0A[Type][Status]<SUB>[Id]<FS>[Id]...	Returns stored config identifiers.
T0B Delete Config	[ConfigType][Ids...].	T0C[Type][Status]<SUB>[PerIdStatus]...	Deletes stored config and flags EMV reload.
T11 Application Selection	No payload. Optional display context may be set by ZA.	Async T12[Result][AID/Application data] or T124 fallback/MSR reason.	Starts contact app selection only. Shows insert-card UI and app-selection list if multiple applications. Remembers selected app for following T15.
T13 Application Selection Alias	Same as T11.	Async T12.	Compatibility alias of T11 in this implementation.
T15 Start Contact Transaction	Transaction request fields plus optional <SUB>[PinScheme]. Default PinScheme 0 MK/SK; 1 DUKPT.	Async T16[Status/Result].	Runs contact EMV flow. Handles online PIN by secure PIN entry and returns PIN data tags through configured data objects.
T17 Complete Online Authorization	Host response data: approval/decline and optional issuer data/TLV.	On failure: T16 failure. On success, transaction completion result is sent asynchronously by EMV callback.	Completes second generation AC. Approved sale can trigger brand sensory. Declines show declined UI.
T19 Add Issuer Script	Issuer script TLV/template payload.	T20[Status].	Stores issuer script to be applied during completion.
T1C Cancel Contact Transaction	No payload.	ACK/no frame.	Cancels active EMV flow, clears transaction display, and returns to idle/cancel state.
T1D Overwrite Runtime Data	Runtime EMV TLV data.	T1E[Status].	Overwrites transaction/kernel data objects for current flow.
T21 Query EMV Data	[DOL/tag list].	T22[TLV data].	Returns requested transaction tags from kernel/cache.
T23 Clear Transaction	No payload.	ACK/no frame.	Clears EMV batch/transaction log state.

Command	Request fields	Response fields	Flow and notes
Log			
T25 Get Batch Data	No payload.	T26[BatchData] or T260.	Returns next batch record from EMV transaction log.
T27 Get Online Authorization Data	No payload.	T28[TLV data up to response packet limit].	Returns online-request data. Payload is truncated to configured max packet size to protect older host clients.
T29 Get Reversal Data	No payload.	T2A[TLV data].	Returns cached reversal data for the last transaction if available.
T2B Get Online Data Packet	No payload.	T2C10<SUB>[TLV data].	Single final-packet form of online authorization data.
T2H Get Transaction Data	[Type] 1 online, 3 batch, 4 reversal.	T2I10<SUB>[Data] or T2I3 invalid type.	Multipacket-compatible query currently returned as a single final packet.
T31 Contact Card Data Read	No payload.	Async T12 selection/card-read result.	Non-financial contact read flow. Used with T3C/T27/T21 to extract card data.
T33 Fast Contact Read	Transaction request payload.	Async T34[Status][Result]<SUB>[Control][Seq][ARQData].	Fast contact flow. T34 control 0 means more data, 1 last packet. Current output is single final packet.
T34 Fast Contact Ack/Response Id	Host ack compatibility frame.	ACK/no frame when host sends T34.	Device sends T34 after T33. Host should ACK the frame at transport level.
T35 Non-EMV Contact Transaction	Transaction request payload.	Async T16.	Starts contact transaction flow used for card-data/non-EMV read cases.
T37 Contact PIN Management	PIN-management payload.	T38[Failure].	Currently unsupported; returns fail response.
T38 PIN Management Response Id	Host response compatibility frame.	T3812.	Compatibility response for unsupported PIN management path.
T3C Force Complete Non-EMV	No payload.	T3C0[Result][FinancialNeed] or failure.	Completes non-EMV card-data read as online request/financial-needed result.

4.8 EMV Contactless Configuration and Contactless Transactions

Command	Request fields	Response fields	Flow and notes
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Command	Request fields	Response fields	Flow and notes
T51 Load Contactless Terminal Config	Configuration packet/TLV payload.	T520 success or failure.	Stores shared terminal configuration and flags reload.
T53 Load Contactless CAPK	[Sequence][CAPK packet].	T54[Sequence][Status].	Stores CAPK only if not already present, because SDK CAPKs are shared between contact and contactless.
T55 Load Contactless Application Config	PCD application packet/TLV payload.	T560 success or failure.	Stores contactless AID/application config.
T59 Query Contactless Config IDs	[ConfigType].	T5A[Type][Status]<SUB>[Id]...	Lists stored contactless config identifiers.
T5B Delete Contactless Config	[ConfigType][Ids...].	T5C[Type][Status]<SUB>[PerIdStatus]...	Deletes stored contactless config and flags reload.
T5D Housekeeping	[ConfigType?].	T5E<SUB>10 success or T5E<SUB>1F failure.	Compatibility housekeeping command.
T5F Load DRL	DRL packet payload.	T5G[Status].	Stores contactless dynamic reader limit configuration.
T5H Delete DRL	No payload.	T5I[Status].	Deletes stored DRL configuration.
T61 Start Contactless Transaction	Transaction request fields plus optional <SUB>[PinScheme]. Default 0 MK/SK; 1 DUKPT.	Async T62[Status][Result].	Shows tap-card UI, soft LEDs, and beep cues. Online PIN callback starts secure PIN entry using the provided session-key scheme.
T63 Query Contactless Data	[ReadyControl?][DOL/tag list].	T64F no data or T6411<SUB>[TLV data].	Returns requested contactless EMV tags.
T65 Query Contactless Online Data	No payload.	T66F no data or T6611<FS>[TLV data].	Returns online authorization data for contactless flow.
T67 Query Payment Scheme	No payload.	T68[SchemeCode].	Maps selected AID to scheme code: Mastercard 22, Visa 32, Amex 42, JCB 53, Discover 62, UnionPay-like 72, F unknown.
T6C Cancel Contactless Transaction	No payload.	ACK/no frame.	Cancels active RF EMV flow and clears transaction display.
T71 Complete Contactless Authorization	Host authorization response/TLV.	On failure: T62 failure. On success, completion is produced by EMV callback.	Completes online contactless transaction.

Command	Request fields	Response fields	Flow and notes
T72 Contactless Compatibility Ack	No payload.	ACK/no frame.	Accepted no-op used by some host flows.
T73 Contactless Issuer Script	Issuer script payload.	T74[Status].	Stores issuer script data for contactless completion.
T75 Revocation List Setup	List payload.	T760 or T7612.	Accepts non-empty list payload after optional SUB/spacing cleanup.
T77 Exception List Setup	List payload.	T780 or T7812.	Accepts non-empty list payload after optional SUB/spacing cleanup.
T81 Start Contactless EMV	Transaction request payload.	Async T62.	Compatibility alias for starting contactless EMV transaction without ZA display context.

5. Standard Flows

Host Flow Examples

Flow	Sequence	Device behavior
Basic swipe sale	ZA optional -> Q1 -> 81 -> Q2 or 72	Shows amount/card prompt if ZA is present, arms MSR, sends track data, then returns to idle when completed.
Multi-interface card detect	ZA optional -> QK -> QK1/2/3 -> Q1 or T11/T61 path	Shows swipe/insert/tap screen. Swipe and RF track data are cached for the next read command when available.
Contact EMV sale	T11 -> T12 -> T15 -> T16(A1) -> T27/T2B -> host auth -> T17 -> T16(final)	T11 selects app. T15 runs EMV. Online PIN is collected via secure PIN entry when requested. Final result drives approval/decline UI.
Contactless EMV sale	ZA optional -> T61 -> T62(A1 or final) -> T65/T63 -> T71 if online	Shows tap prompt, soft LEDs, and beeps. Online PIN is requested through the selected scheme from T61 payload.
Non-financial card data read	T31 -> T12 -> T3C -> T27/T21	Reads application/card data without completing a normal sale authorization.
Data entry	Z2/Z3 prompt -> Z42 or Z50 -> Z43/Z51	Displays prompt, waits for keypad/touch input, returns entered key/text, or EOT on cancel/timeout.

6. Prompt Appendix

Fixed Prompt Identifiers

Prompt type	Ids	Usage
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Prompt type	Ids	Usage
Data entry	001 ACCOUNT NUMBER; 003 ENTER CUST ID; 004 ENTER AMOUNT; 015 ENTER CASH BACK; 024 ENTER; 100 ACCOUNT NUMBER; 114 CUSTOMER REF; 115 CUSTOMER REF NO.; 126 ENTER BADGE #; 129 ENTER CASH BACK; 134 ENTER CUST REF; 200 ENTER OTP; 201 ENTER VERIFICATION CODE; 202 ENTER CVV2; 203 ENTER SMS OTP; 204 ENTER AUTHENTICATOR CODE; 205 ENTER TEMPORARY PASSWORD.	Use with Z2/Z3 fixed-prompt mode and then Z42/Z50.
PIN	001 ENTER PIN; 002 REENTER PIN; 006 ENTER NEW PIN; 007 CONFIRM NEW PIN; 024 ENTER.	Use with PIN prompt commands or Z2/Z3 PIN fixed-prompt mode.

7. Operational Notes

- Press ENTER + 1 on the CT20P keypad to open the settings menu. Serial settings, Android settings, Wi-Fi settings, and exit actions are managed there.
- The app hides system navigation/control bars during normal operation and restores them when exiting to Android.
- Settings password protection can be enabled at build time. Android settings and exit are protected when enabled.
- Keys are injected into secure hardware; local metadata is required to know intended usage after app reinstall. If metadata is absent, PIN entry fails safely and instructs re-injection.
- Contact/contactless EMV configuration is loaded into the SDK at application start and whenever configuration changes mark reload as required.
- Build variables control protocol logging, EMV debug logging, and brand sensory display.

8. Revision History

Revisions

Revision	Date	Changes
1.1	2026-07-03	Expanded command dictionary with per-command request fields, response fields, flows, and implementation notes.
1.0	2026-07-03	Initial NEXGO Android Smart PinPAD manual.